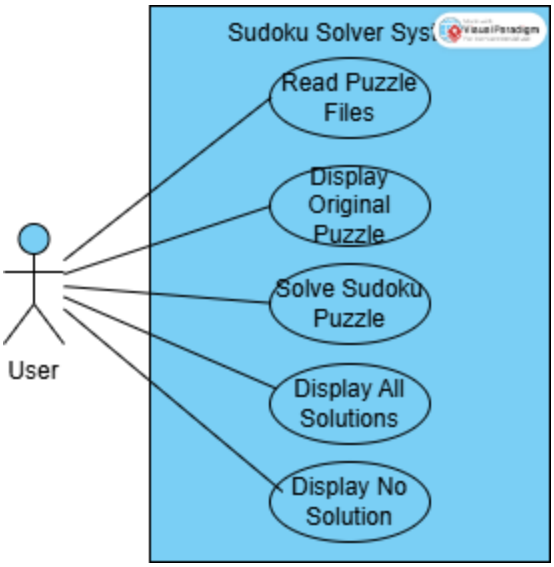


1. Use Case Name
Solve Sudoku Puzzle

2. Brief Description
This use case shows how the Sudoku Solver works. The program reads Sudoku puzzles from text files, solves them using backtracking, and displays the answers. It also checks that the numbers follow the Sudoku rules.

3. Step-by-Step Flow of Events

1. The user runs the Sudoku Solver program.
2. The program opens and reads a Sudoku puzzle file.
3. It displays the original puzzle.
4. The program checks if the given numbers are valid.
5. It finds an empty space and tries numbers from 1 to 9.
6. If a number fits, it moves on to the next empty space.
7. If a number doesn't work, it goes back and tries a different one.
8. Once the puzzle is solved, it displays the solution.
9. If there is more than one solution, it shows all of them.
10. If the puzzle cannot be solved, it displays "No solution found."
11. The program does the same thing for all five puzzle files.

4. Detailed Description

Actor: User

System: Sudoku Solver System

Preconditions: The Sudoku puzzle files need to be available for the program to read.

Trigger: The user starts the program.

Main Flow: The program reads each Sudoku puzzle and shows the original grid. It then tries different numbers in the empty spaces while checking the Sudoku rules. If a number doesn't work, it goes back and tries another one until it finds a solution.

Alternative Flow: If the puzzle has more than one solution, the program shows all of them. If no solution is possible, it displays "No solution found." If a file is missing or has incorrect information, the program shows an error message.

Postconditions: At the end, the program displays the original puzzle along with its solution or a message saying that no solution was found.